MINDFULLY YOURS

How the AI produces a reply

The flow from a user's message, through your Knowledge Base and therapist conversations, the database and the AI, to the reply the user receives.

Prepared for the Mindfully Yours clinical and product team | Version 1.0

The short answer

1. **Your therapists' conversations come first.** Every message is compared, by meaning, with the approved therapist replies in the library. When one fits, the user gets *your therapist's reply*, as written or lightly reworded. The AI language model writes a reply on its own only when nothing in the library fits, and then it stays short and careful.
2. **Safety is checked first, every time.** A message that matches a red flag gets your team's approved script, word for word. No AI writing is involved.
3. **One PostgreSQL database does two jobs.** The ordinary tables (the RDS) hold the text, rules and scripts; the vector search (pgvector) inside the same database finds the closest match by meaning. They are not two separate systems.
4. **Users in other languages get the same treatment**: the message is translated for the safety check and the search, and the reply comes back in their own language.

Therapists first
approved therapist replies are searched before anything else

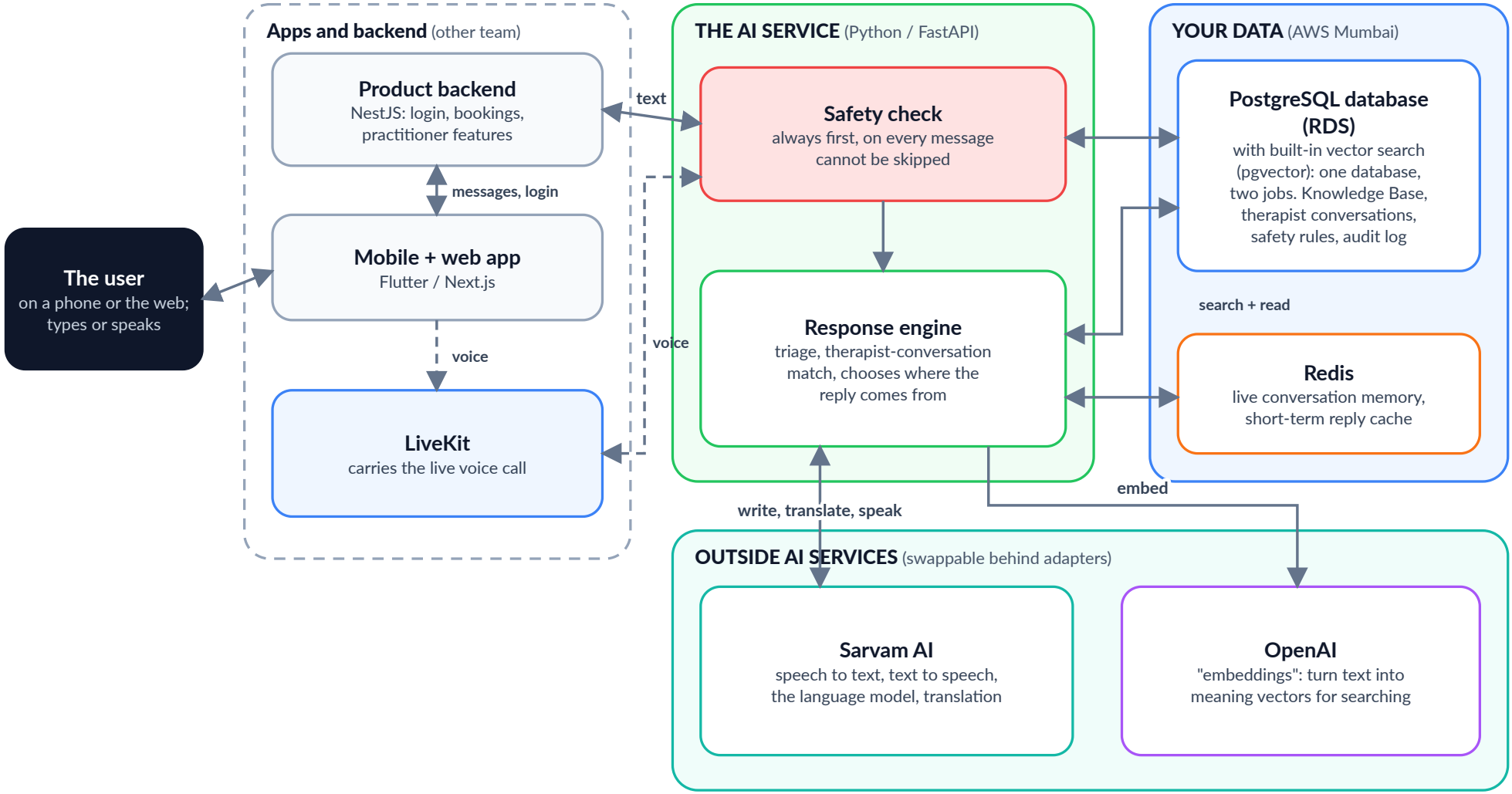
Safety first
every message is checked against the red flags

Your language
users in other languages get the same care

One database
PostgreSQL holds both the tables and the vector search

This document describes the approach: how a user's message is understood, how the Knowledge Base and the therapist conversations are searched, and how the reply is chosen and delivered.

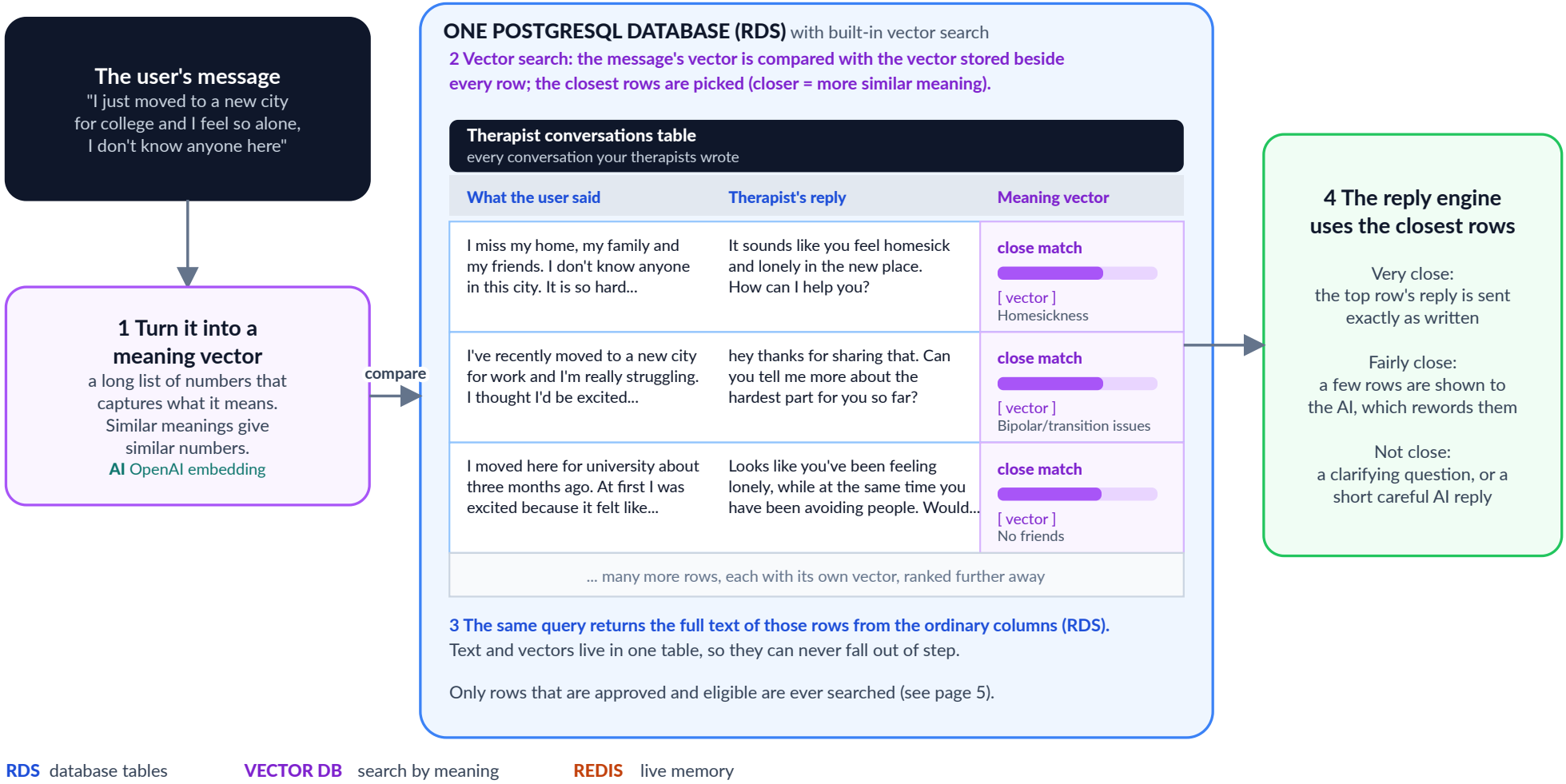
The system at a glance



RDS database tables **VECTOR DB** search by meaning **REDIS** live memory **AI** outside AI service

Hosting: the architecture places the database and services in AWS Mumbai (India) for data residency. The outside AI services can be swapped behind adapters, with your written consent.

RDS and the vector database, working together on one question



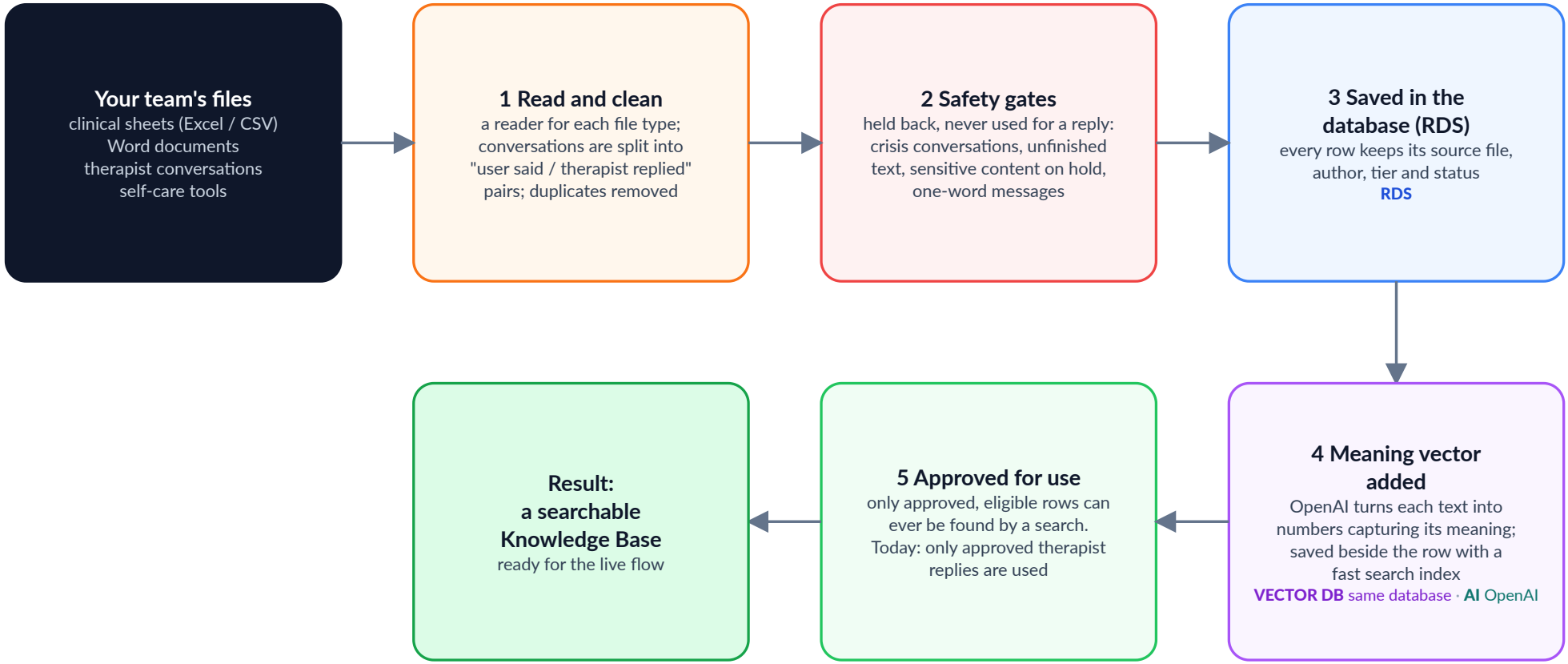
This is a real example from the running system. The user's message is turned into a meaning vector, compared with the vector stored beside each therapist conversation, and the closest rows come back together with their full text. The three closest rows are all about moving to a new place and struggling with it, even though none uses the same words.

What is stored where

What it is	Where it lives	Used by
Safety red flags and approved scripts (with translated copies)	RDS VECTOR	Safety check (step 7)
Signals people express, and symptoms	RDS VECTOR	Triage (step 6)
Patterns and how to keep probing	RDS VECTOR	Triage; guidance for the AI
Rules: when something counts as Tier 1, 2 or 3	RDS	Triage (read by code, not searched)
Therapist conversations (user said / therapist replied); only approved ones are searched	RDS VECTOR	Therapist match (steps 5, 8)
Therapist clarifying questions (the general ones are used)	RDS VECTOR	When nothing matches (step 8)
Self-care advice for everyday situations	RDS VECTOR	Guidance for the AI (step 8)
Welcome messages	RDS	Start of a conversation
Audit trail of every reply	RDS	Record the turn (step 11)
Live conversation memory; short-term reply cache	REDIS	Steps 2 and 11

RDS = stored as ordinary rows of text and rules. **VECTOR** = also given a meaning vector so it can be searched by meaning. **REDIS** = fast short-lived memory, not permanent storage.

How the Knowledge Base gets into the database



RDS database tables **VECTOR DB** search by meaning **REDIS** live memory **AI** outside AI service

Re-run whenever your team adds or changes files: it only adds what is new.

Your team's files are read once, checked, saved, given meaning vectors and approved. Rows that could be unsafe or unfinished are kept in the database but are never searchable (no vector, not approved): crisis conversations (the safety check owns those), unfinished text, content on governance hold, and one-word messages that would match anything. The loads so far cover the clinical sheets, the question banks, the self-care tools and the therapist conversations.

Where does the reply come from?

There are five possible sources for a reply. They are tried in this order, and the first one that applies is used. The closer the match, the more the reply is purely your therapists' words.

1	**Approved safety script, word for word** A red flag is detected in the message	Your team's words only
2	**The therapist's reply, exactly as written** A very close match in the library (a strong match)	Your therapists' words only. The AI is not used.
3	**The therapists' replies, lightly reworded** A fair match (a partial match)	The AI sees a few close therapist examples and may add nothing new
4	**A therapist-written clarifying question** Nothing matches, and nothing else can guide the reply	Your therapists' words only
5	**A short, careful AI reply** Still nothing that fits	AI language model, with self-care advice if found; suggests a practitioner when unsure

What "close" means

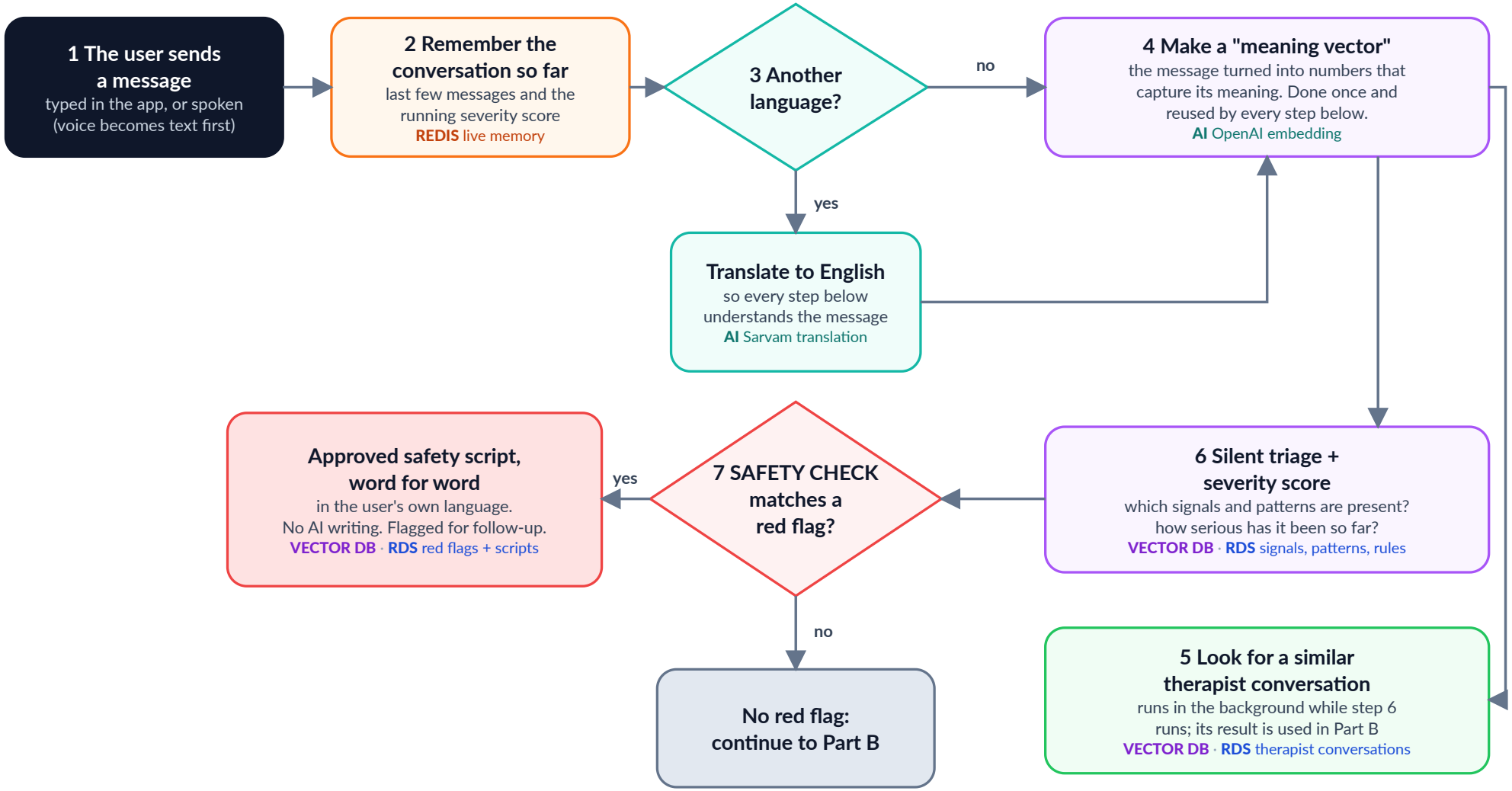
Every message and every therapist conversation is turned into a meaning vector. "How close" is a similarity score between the two. The cut-offs between strong, partial and none are **provisional starting values**, set conservatively so that "exactly as written" is only used for very clear matches. They are meant to be set with your therapists, using a sample of messages they mark as "fits" or "does not fit".

Always true

- The safety check runs before everything else.
- The AI never decides whether something is a crisis.
- Every reply is recorded with which of the five sources produced it, so the share coming from your therapists can be reported.

THE FLOW CHART

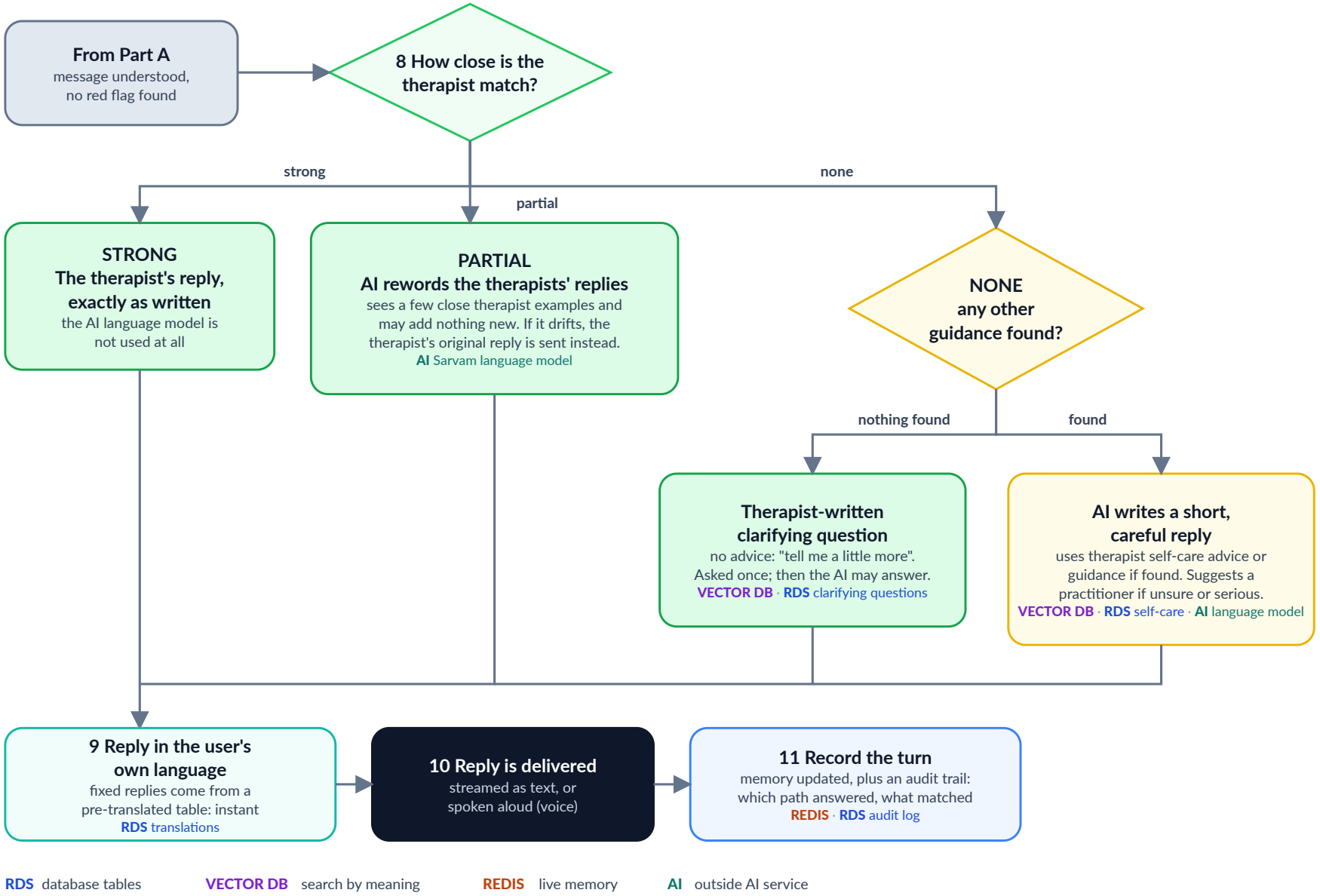
Step by step, part A: message in, safety checked



RDS database tables **VECTOR DB** search by meaning **REDIS** live memory **AI** outside AI service

THE FLOW CHART

Step by step, part B: choosing the reply and sending it



What the AI language model sees (the RAG step)

RAG means *retrieve, then generate*: first the system **retrieves** the relevant material from the database; only then is the AI asked to **write** a reply using that material. The AI is never asked to work from memory alone.

Always given to the AI

- The persona and tone (warm, calm, non-clinical)
- The language to reply in
- A short checklist of what to quietly notice (the Clinical Markers)
- The last few messages of this conversation
- The user's message

Plus one kind of retrieved material

- **Close therapist examples** (a few) when the match is partial
- **Self-care advice** found by searching for the user's situation
- **Therapist probing guidance** for the closest pattern
- A gentle **booking suggestion**, once concern has persisted and its duration and impact are clear
- A therapist **follow-up question** for a possible concern

The AI is never asked to

- decide whether a message is a crisis (the safety check does that)
- add advice to a matched therapist reply
- work from other users' conversations
- diagnose, or name a clinical condition to the user

Checked before the user sees it

When the AI rewords therapist replies, the result is checked: it must reuse the therapists' own wording, add no numbers or links, and stay about as short. If it drifts, the therapist's original reply is sent instead.

A real example of what the AI is shown (shortened), for a partial match

Real examples from our therapist team, closest first:
1. When someone said: "I miss my home, my family and my friends. I don't know anyone in this city..."
   the therapist replied: "It sounds like you feel homesick and lonely in the new place. How can I help you?"
2. When someone said: "I've recently moved to a new city for work and I'm really struggling..."
   the therapist replied: "hey thanks for sharing that. Can you tell me more about the hardest part for you so far?"
Write ONE reply, using ONLY the ideas, wording and questions already in the therapist replies. Do NOT add advice or facts that are not in them.
The user just said: "I just moved to a new city for college and I feel so alone..."
The AI wrote: "It sounds like you feel alone in the new place. How can I help you?"

Worked examples from the running system

What the user said	How close	Where the reply came from	What the user received
"Hi, I just moved away from my home last month, and things have not been easy ever since."	Very close (strong)	Therapist's reply, as written	"Moving away from home can be quite tough. What about this move has been the most difficult for you?"
"I just moved to a new city for college and I feel so alone, I don't know anyone here"	Fair (partial)	Therapists' replies, lightly reworded	"It sounds like you feel alone in the new place. How can I help you?"
"I just don't really know how to put this into words"	None	Therapist clarifying question	"I'm not completely sure I understand what you mean. Could you tell me a little more?"
"my teenage son won't talk to me anymore"	None	Short careful AI reply	"Oh, that sounds really heavy, and I can hear how much it hurts you. It takes so much patience to watch someone you love pull away..."

These are test messages run through the real pipeline (not user data). Messages in other languages take slightly longer because of the translation step.